Two orientations of the fermionic force
The projected second memory coefficient in energy space

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Abstract

The output-rank deficit in the second memory coefficient is asymptotic to the average of the annihilation and creation force intensities. We prove this reduction using an exact Casimir identity and overlap bounds. The two force orientations have a positive matrix of q^2 probe measures. An endpoint certificate proves that their cross measures with the fourth-chaos residual vanish at every fixed energy order. The second memory coefficient, after the previously specified eight-direction projection, therefore has a positive sum of the existing q^4 and q^2 Mehler kernels.

1 State, conventions, and result

Use standard CAR, the full overlapping complex Gaussian p-body Hamiltonian, fixed charge $m = N/2$, and the normalized maximally mixed trace. All inner products are annealed: $\langle X, Y\rangle = \mathbb{E}\,\mathrm{tr}_m\, X^\dagger Y$. Set $\hbar = 1$ and

$$h = H_m, \quad \lambda_N = p^2/N \longrightarrow \lambda > 0, \quad q = e^{-4\lambda}, \quad C_N = \mathbb{E}\,\mathrm{tr}_m\, h^2 \longrightarrow C = J_s^2/\lambda. \tag{1}$$

The variance convention and the same-coupling charge lifts are those of the preceding notes. W_j denotes ordinary Gaussian-coordinate chaos, not the spectral q-Wick polynomial of Ref. [1]. Define $d_i = [H, c_i]$ on full Fock space and restrict neutral operators to charge m:

$$A_- = \sum_i d_i^\dagger d_i, \qquad A_+ = \sum_i d_i d_i^\dagger, \quad O_\pm = \mathsf{W}_2 A_\pm, \quad B = \mathsf{W}_2 h^2. \tag{2}$$

Let $\widehat{\Pi}_N$ be the actual finite-N orthogonal projection onto $\mathrm{span}\{I, B, O_-, J_h F, E_0, M_0, S_0, Z_0\}$, as in Ref. [3]. For $X_N = (1 - \widehat{\Pi}_N) m Z_2$, our result is

$$\boxed{d\nu_X = V_4\, d\mu_{q^4} + V_2\, d\mu_{q^2}, \qquad C_X(t) = V_4 G_{q^4}(t) + V_2 G_{q^2}(t).} \tag{3}$$

Here μ_r is the normalized, existing two-energy probe measure, $G_r(0) = 1$, and both weights are positive. This is the full two-point spectrum of this specified projected coefficient; the finite-time memory operator remains a separate problem.

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2 An exact bridge from output rank to force orientation

Let $Q = \sum_i c_i^\dagger c_i$ and define on neutral full-Fock operators

$$DX = QX - \sum_i c_i^\dagger X c_i, \qquad \overline{D}X = (N-Q)X - \sum_i c_i X c_i^\dagger. \tag{4}$$

For the neutral quadratic Casimir $\mathcal{C}_x$, with eigenvalues $x_a = a(N+1-a)$ on rank a, CAR gives at half filling

$$\mathcal{C}_x X = (m+1)DX + m\overline{D}X - D\overline{D}X, \qquad DH = pH. \tag{5}$$

The compositions on the right are formed before charge restriction. Write $h = \sum_a h_a$, $u = p - a$, and

$$D_{\rm in} h = \sum_a u h_a, \qquad E_{\rm in} = \sum_a \frac{u(u-1)}{m-p+1} h_a. \tag{6}$$

Equation (5) implies the exact relation

$$(p - \overline{D})h = 2D_{\rm in}h + E_{\rm in}. \tag{7}$$

Indeed $\overline{D}h_a = (x_a - pm)h_a/(m-p+1)$, and $(x_p - x_a)/(m-p+1) = 2u + u(u-1)/(m-p+1)$.

Let $Z^+_{ab;r}$ be the rank-r component of $h_a h_b + h_b h_a$ for $a < b$, and of h_a^2 for $a = b$. Define the actual deficit and its Casimir proxy by

$$\mathcal{S}_N = \sum_{a \le b, r} (a + b - r) \mathsf{W}_2 Z^+_{ab;r}, \tag{8}$$

$$\widetilde{\mathcal{S}}_N = \sum_{a \le b, r} \frac{x_{a+b} - x_r}{N} \mathsf{W}_2 Z^+_{ab;r}. \tag{9}$$

The earlier positive rank-moment bounds give $\|\mathcal{S}_N - \widetilde{\mathcal{S}}_N\|_2 = O(p^{-1})$. Introduce full-Fock operators and then restrict:

$$\overline{A} = (2p - \overline{D})H^2 = A_+ + \{H, (p - \overline{D})H\}, \tag{10}$$

$$E_{\rm dbl} = (2p - D)\overline{A}, \qquad U_{\rm in,2} = \{h, D_{\rm in}^2 h\} + 2(D_{\rm in}h)^2. \tag{11}$$

Substitution into (5), with $A_- = (2p - D)H^2$, gives an exact finite-N identity:

$$\begin{aligned} N\left(\widetilde{\mathcal{S}}_N - \frac{O_- + O_+}{2}\right) = (1-2p)O_- - 2pO_+ - \mathsf{W}_2\{h, D_{\rm in}h\} \\ - \mathsf{W}_2 U_{\rm in,2} + \mathsf{W}_2 E_{\rm dbl} + (m-2p)\mathsf{W}_2\{h, E_{\rm in}\}. \end{aligned} \tag{12}$$

It follows either by collecting $x_a + x_b - 2ab - x_r$, or by applying the Casimir directly to h^2. No division by a possibly vanishing rank product is used.

3 Controlling the bridge error

The only new bound needed in (12) is the normal/reverse double mark. We give its local reason explicitly. On a neutral word, the sitewise maps in $D = \sum_i D_i$, $\overline{D} = \sum_i \overline{D}_i$ have the table below; fermion signs in odd local factors are fixed by the original global ordering. Put $n = c^\dagger c$.

local factor	I	c	$c^\dagger$	n	$1-n$
D_i	0	0	$c^\dagger$	n	$-n$
$\overline{D}_i$	0	c	0	$n-1$	$1-n$

Consider one two-slot monomial in H^2, before summing indices. Let a_i, b_i count its original creation and annihilation factors at site i. Then $\sum_i a_i = \sum_i b_i = 2p$ and

$$(2p - D)(2p - \overline{D}) = \left[\sum_i (a_i - D_i)\right]\left[\sum_j (b_j - \overline{D}_j)\right]. \tag{13}$$

At a site used by just one of the four original index sets, both deficits annihilate its local factor. The remaining sites number at most K, the sum of pairwise index-set overlaps. Each local map has a bounded coefficient norm. Thus one deficit has coefficient bound $K_0 K$, and two deficits have bound $K_0^2 K^2$ on the overlap sites. A prior deficit does not modify a singleton site, so the second operation creates no hidden factor of p there.

In the Gaussian second chaos, internal contractions of a two-slot block are absent. For each fixed Wick graph, its distinct p-sets have uniformly bounded $\mathbb{E}e^{aK}$ for every fixed a at fixed p^2/N. This follows from the hypergeometric factorial-moment bound for each pair and Hölder for the finite number of pairs. After local occupation expansion, absolute trace coefficients are bounded by $K_1(1+K)^j e^{K_2 K}$ times the singleton normalization. The same argument treats any fixed number of blocks and marks.

For the half-filled trace, the exact local/full-Fock ratio is

$$R_N(L,t) = 2^L \frac{(m)_t^{\downarrow}(m)_{L-t}^{\downarrow}}{(N)_L^{\downarrow}} \le e^{K_3 p^2/N}. \tag{14}$$

On fixed overlap windows the balanced singletons give $|2t - L| \le K_4 K$, hence $R_N \to 1$. The exponential overlap bound controls the complementary window, as in Ref. [3]. Consequently $\|O_\pm\|_2$ and $\|\mathsf{W}_2 E_{\mathrm{dbl}}\|_2$ are bounded. The earlier factorial input-rank tails also bound every fixed moment of $D_{\mathrm{in}}^j h$, while those of E_{in} are $O(N^{-1})$ in norm. Gaussian fourth-moment bounds and Hölder therefore bound the products in (12). We obtain

$$\boxed{\mathcal{S}_N = \tfrac{1}{2}(O_- + O_+) + O_{L^2}(p^{-1}).} \tag{15}$$

The limits and constants here keep $\lambda > 0$ fixed.

4 The oriented two-energy spectral matrix

For two marked pairs (a, b) and (c, d) in a Wick word, let π be the Gaussian pairing. The local determinant calculation gives the same single-mark coefficient for either orientation, $\alpha = 2 - 4\,\mathbf{1}_{\text{incident crossing}}$. The mixed coefficient is

$$\delta = \begin{cases} 4\,\mathbf{1}_{\pi(a)=d,\,\pi(b)=c}, & \text{equal orientations,} \\ -4\,\mathbf{1}_{\pi(a)=c,\,\pi(b)=d}, & \text{opposite orientations.} \end{cases} \tag{16}$$

The determinant entry for a normal mark (x, y) changes by -2 at $(\pi(j), i) = (x, y)$; for a reverse mark it changes by $+2$ at $(\pi(j), i) = (y, x)$. These statements hold before any spectrum is postulated. The two-mark weight is $q^{\mathrm{cr}}(\lambda\delta + \lambda^2\alpha\beta)$ times the appropriate power of C.

For arbitrary nonnegative a, b, put $n = a + b$. In unit-variance energy variables let P_{ab} be the mixed moment of μ_{q^2}, let m_j be the q-Gaussian moment, and put

$$u_q = \frac{1-q}{1+q}, \qquad S_n = m_{n+4} - 2m_{n+2} + m_n. \tag{17}$$

Swapping the two adjacent endpoints of each force insertion changes only their mutual crossing. The usual four-member exchange orbits give u_q^2; the completely connected double bridge has two members. For opposite orientations only its crossing member contributes to δ. Thus for all two-arc words

$$\sum q^{\mathrm{cr}}\delta_{--} = 4P_{ab}, \qquad \sum q^{\mathrm{cr}}\delta_{-+} = -4qP_{ab}, \tag{18}$$

$$\sum q^{\mathrm{cr}}\alpha\beta = 4u_q^2 S_n + \frac{16q}{1+q}P_{ab}, \qquad \sum q^{\mathrm{cr}}\alpha = 2u_q S_n. \tag{19}$$

The last identity also fixes both crosses with the conserved B. Set $c = 2\lambda u_q$ and $Y_\pm = O_\pm - cB$. Subtracting this common background yields the two-energy matrix

$$d\nu_{(Y_-,Y_+)} = \begin{pmatrix} w & x \\ x & w \end{pmatrix} d\mu_{q^2}, \qquad \begin{aligned} w &= 4C^2\lambda + \frac{16C^2\lambda^2 q}{1+q}, \\ x &= -4C^2\lambda q + \frac{16C^2\lambda^2 q}{1+q}. \end{aligned} \tag{20}$$

Both eigenvalues are positive:

$$w + x = 4C^2\lambda(1-q) + \frac{32C^2\lambda^2 q}{1+q}, \qquad w - x = 4C^2\lambda(1+q). \tag{21}$$

The symmetric and antisymmetric force combinations consequently have the same normalized two-point kernel but different strengths.

5 The projected deficit and its cross with the quartic field

Put $v = 2\lambda(1-q) + 16\lambda^2 q/(1+q)$, $d = 2\lambda(1+q)$, and $\rho = v/(v+d)$. The preceding equal-time Gram and its nondegenerate finite-N limit give, for $\Pi_N = \mathrm{proj}_{\{I,B,O_-,J_hF\}}$,

$$T_N = (1-\Pi_N)\mathcal{S}_N = \frac{Y_- + Y_+}{2} - \rho Y_- + o_{L^2}(1). \tag{22}$$

The coefficient of the additional J_hF direction tends to zero; its norm is bounded and its limiting Gram is nondegenerate. No unproved energy-resolved F cross is needed for this replacement. Equation (20) now gives

$$d\nu_T = V_{\mathrm{new}}\, d\mu_{q^2}, \qquad V_{\mathrm{new}} = \frac{C^2 v d}{v+d} = \frac{C^2\lambda(1+q)(1-q^2+8\lambda q)}{1+q+4\lambda q}. \tag{23}$$

The remaining issue is the cross with Q_4. To make the finite certificate reproducible, define

$$d_3 = q^2 - q + 1, \quad d_5 = 1 + q + q^2 + q^3 + q^4, \quad D_4 = (1+q^2)d_3 d_5. \tag{24}$$

In the four-slot mark order $01, 02, 03, 12, 13, 23$, the numerator weights for Q_4 are

$$\begin{aligned} \boldsymbol{w}_4 = (&q^3(1+q^2), -q(1+q^2)d_5, D_4, \\ &q^2(1+q+q^2)(1+q^2), -q(1+q^2)d_5, q^3(1+q^2)). \end{aligned} \tag{25}$$

The unmarked numerator is $2\lambda(q-1)^3(1+q)(1+q+q^2)d_3$. For Y_+, the marked and unmarked numerators are $1+q$ and $-2\lambda(1-q)$, with denominator $1+q$.

Pair a four-slot block with a two-slot block. If they share $k = 0, 1, 2$ bridges, the remaining $6-2k$ endpoints have an A/B color word and one split between the two exterior arcs. There are

$$\sum_{k=0}^{2} \binom{6-2k}{4-k}(7-2k) = 128 \tag{26}$$

templates, each with $4!2!/k!$ pairings. Substituting the opposite-orientation rule (16), all 128 coefficients of λ and λ^2 vanish as integer polynomials in q. The certificate retains all 6072 words' template aggregates. Arbitrary spectator chords contribute a common factor on each endpoint orbit: continuous block permutations preserve crossings with all exterior chords except the explicitly retained endpoints. Thus this finite classification proves cancellation for arbitrary fixed two-arc energy degrees, not just the nine low moments checked separately. Together with the old normal-orientation certificate,

$$\nu_{Q_4,Y_-} = \nu_{Q_4,Y_+} = 0, \qquad \nu_{Q_4,T} = 0. \tag{27}$$

Equation (20) also gives $\nu_{T,Y_-} = 0$ at every energy, not only after energy integration. Thus $\nu_{X,Y_-} = 0$, including crosses with all fixed left/right energy descendants of Y_-.

6 From moments to the full memory spectrum

For Hermitian fields X_i, the finite-system matrix measure is

$$\nu_{ij,N}(dE, dE') = \mathbb{E}\,\mathrm{tr}_m\left[X_i \Pi_h(dE) X_j \Pi_h(dE')\right]. \tag{28}$$

It is a positive matrix-valued measure, as seen in an energy eigenbasis. The fixed-word overlap bounds give all diagonal energy moments, tightness, and uniform integrability of any fixed moment. Spectral Cauchy–Schwarz bounds every off-diagonal total variation by the diagonal measures. The candidate measures have compact energy support, so their full moment sequence is determinate. Polarization therefore gives joint weak convergence for (Q_4, Y_-, Y_+). In particular (27) is a statement about the entire cross measure, not merely the equal-time inner product.

An L^2 replacement $X - X'$ changes the self measure by total variation at most $\|X - X'\|_2(\|X\|_2 + \|X'\|_2)$, by spectral Cauchy–Schwarz on partitions. Hence (15) and (22) also apply to bounded spectral functions without interchanging an unbounded exponential series and the limit.

Ref. [3] established the actual raw-coefficient reduction

$$(1 - \widehat{\Pi}_N) m Z_2 = Q_{4,N} + 8\lambda_N C_N q_N^2 T_N + o_{L^2}(1). \tag{29}$$

The old fourth-chaos theorem and (27) prove (3), with

$$V_2 = 64 J_s^8 q^4 \frac{(1+q)(1-q^2+8\lambda q)}{\lambda(1+q+4\lambda q)}, \tag{30}$$

$$V_4 = C^4 \frac{4\lambda(1-q)^2(1+q)^2[P(q)+4\lambda q^3 U(q)]}{(1+q^2)d_3 d_5}, \tag{31}$$

$$P(q) = 1 + q + 2q^2 + q^3 + q^4, \qquad U(q) = 2 + q + 2q^2 + q^3 + 2q^4. \tag{32}$$

For completeness, the existing probe kernel [1, 2] is

$$d\mu_r = \rho_q(\theta)\rho_q(\phi) K_r(\theta, \phi)\, d\theta\, d\phi, \tag{33}$$

$$\rho_q(\theta) = \frac{(q;q)_\infty |(e^{2i\theta}; q)_\infty|^2}{2\pi}, \tag{34}$$

$$K_r(\theta, \phi) = \frac{(r^2; q)_\infty}{\prod_{s,t=\pm 1}(r e^{i(s\theta + t\phi)}; q)_\infty}, \tag{35}$$

$$E(\theta) = \frac{2\sqrt{C}\cos\theta}{\sqrt{1-q}}, \qquad G_r(t) = \int e^{it(E-E')}\, d\mu_r. \tag{36}$$

Here $\theta, \phi \in [0, \pi]$ and $(z;q)_\infty = \prod_{j \geq 0}(1 - zq^j)$. For fixed $0 < q < 1$ the measures are positive and absolutely continuous; there is no limiting static atom. Their use here follows the same-coupling composite-field reduction, rather than assuming an independent random probe at the outset.

7 Small deformation, evidence, and continuation

After the double-scaled limit, take $\lambda \downarrow 0$ at fixed string time. The known probe limit gives

$$G_{q^2}(t) \longrightarrow \operatorname{sech}^4(2J_s t), \qquad G_{q^4}(t) \longrightarrow \operatorname{sech}^8(2J_s t), \tag{37}$$

$$V_2 \longrightarrow 1024 J_s^8, \qquad V_4 \sim \frac{768 J_s^8}{5\lambda}, \tag{38}$$

$$\frac{V_2}{V_4 + V_2} = \frac{20}{3}\lambda + o(\lambda), \qquad \frac{C_X(t)}{V_4 + V_2} \longrightarrow \operatorname{sech}^8(2J_s t). \tag{39}$$

Thus the lower-chaos contribution survives with finite absolute strength while its fraction vanishes in this ordered limit. The coefficient $20/3$ describes the weight fraction only; it is not the complete first correction to the time kernel, which also receives the deformation of G_{q^4}.

Executed checks. The local normal/reverse determinant rule was checked in 1688 four-, six-, and eight-slot cases. Direct CAR matrices supplied 640 words and 2560 comparisons, including 115 nonzero double-mark values and three negative values. Five small half-filled systems, each with three deterministic nonzero Hermitian samples, supplied 90 exact bridge identities. Cleared complex-integer entries stayed below 2^{48}; the Casimir implementation uses $2\mathcal{C}_x$.

The new opposite-orientation endpoint certificate has 128 templates and no nonzero polynomial. A separate direct pairing enumeration checks nine (a, b) mixed moments through $a+b = 4$, including the single-mark and double-mark sums. Symbolic checks reproduce the previous deficit Gram and the Schur residual V_{new}, both positive eigenvalues, and the two small-λ coefficients.

For $J_s = 1$, $\lambda = 0.25, 0.1, 0.025$ and $t = 0, 0.25, 0.5, 1$, a symmetric q-Jacobi evolution evaluates both kernels and their mixture. A Taylor bound controls normalized truncation below 10^{-13}; twelve extra Jacobi levels check refinement. At the first two couplings, independent 128- and 192-node Mehler quadrature agrees within 2×10^{-10}. The numerical file retains all weights, values, cutoffs, and errors. Floating rounding and quadrature are refinement checks, not interval certificates. Source hashes and dependency versions are stored alongside each result.

Next physical step. The specified raw mZ_2 projection is now closed at two points. The next tasks are its mixed spectra with earlier memory fields, a uniform central-charge-window extension, and a finite-memory two-parameter remainder. These feed the Susskind singlet/meson program. A chord label 2 or 4 alone does not fix a bulk meson mass or a spatial reconstruction map.

References

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